

Student Depression Dataset Analysis

CSP-571 | Data Preparation and Analysis | Spring 2025

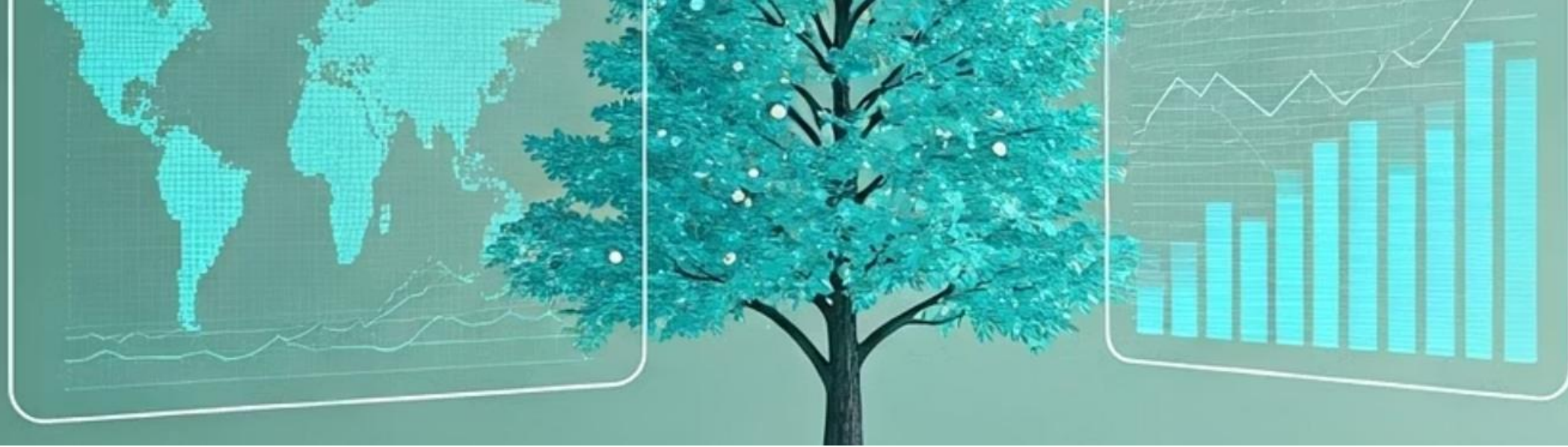
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Research Abstract

Investigates mental health patterns in university students.

Focus on academic, financial, and lifestyle stressors.

Random Forest model achieved 84% accuracy in prediction.

Identifies high-risk profiles for targeted mental health interventions.



Problem Statement

Prevalence

Over 25% of students report depressive symptoms (WHO/ACHA).

Key Stressors

Academic pressure, work hours, financial difficulties, poor sleep.

Critical Need

Develop tools for early identification of depression in students.

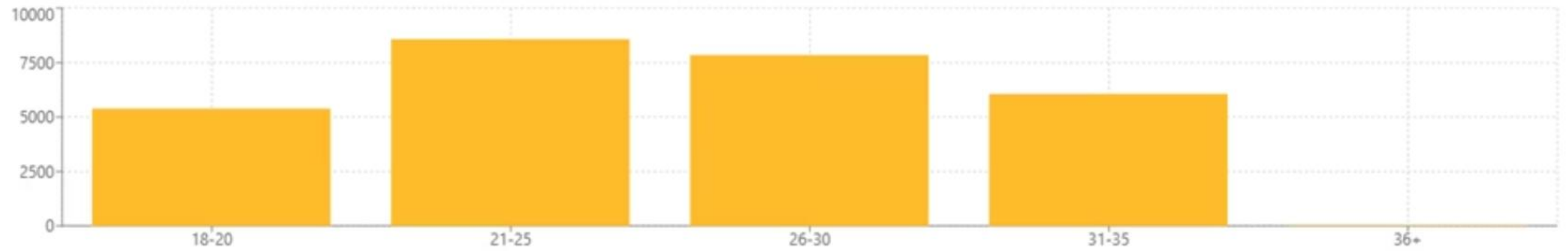


Dataset Overview

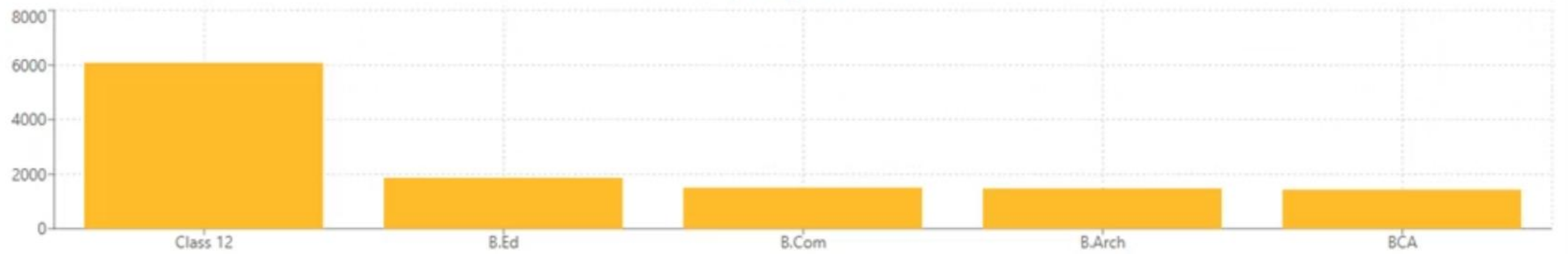
Number of Responses	27,901
Attributes	18 (demographics, CGPA, study habits, stress, suicidal thoughts)
Target Variable	Binary depression diagnosis (0 = No, 1 = Yes)
Class Distribution	26% depressed, 74% non-depressed



Age Distribution



Top 5 Degrees



Project Workflow

1. Data Acquisition & Understanding

2. Data Cleaning & Preprocessing

3. Feature Engineering

4. Exploratory Data Analysis (EDA)

5. Model Development & Training

6. Unsupervised Clustering

7. Interpretation & Insights

Data Cleaning & Preprocessing

Cleaning

Minimal missing data; only 3 missing values in Financial Stress.

Encoding

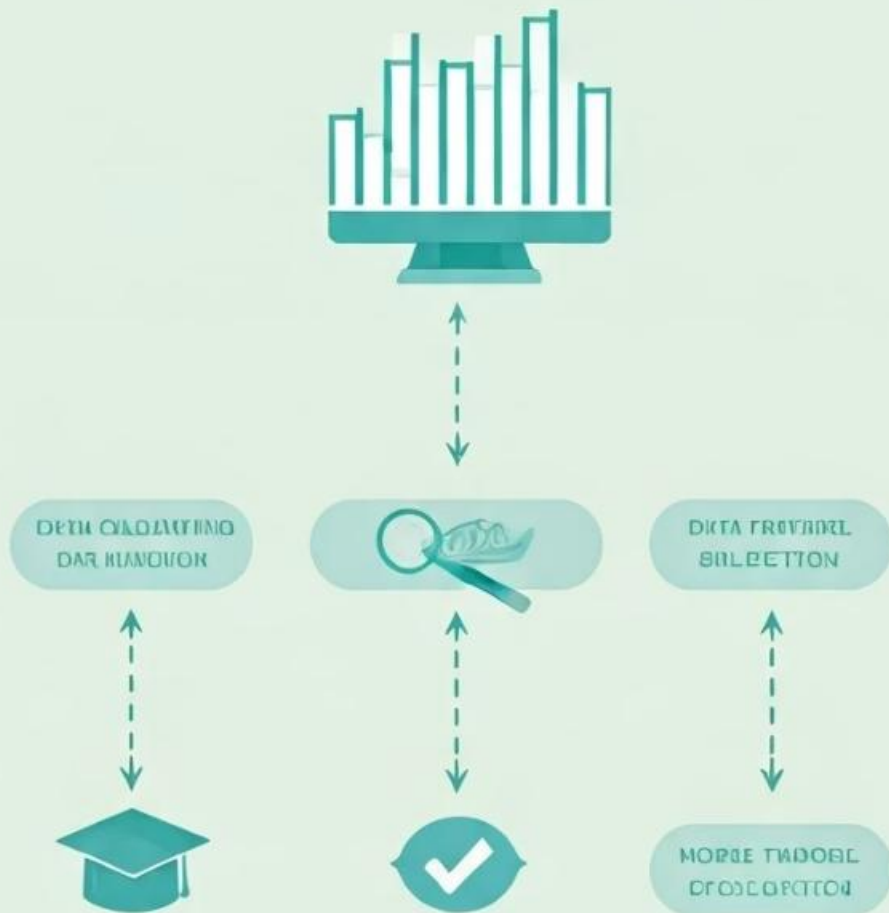
Label encoding for binary and one-hot for multi-category features.

New Feature

Created "Stress Index" combining academic, work, and financial stress.

Normalization

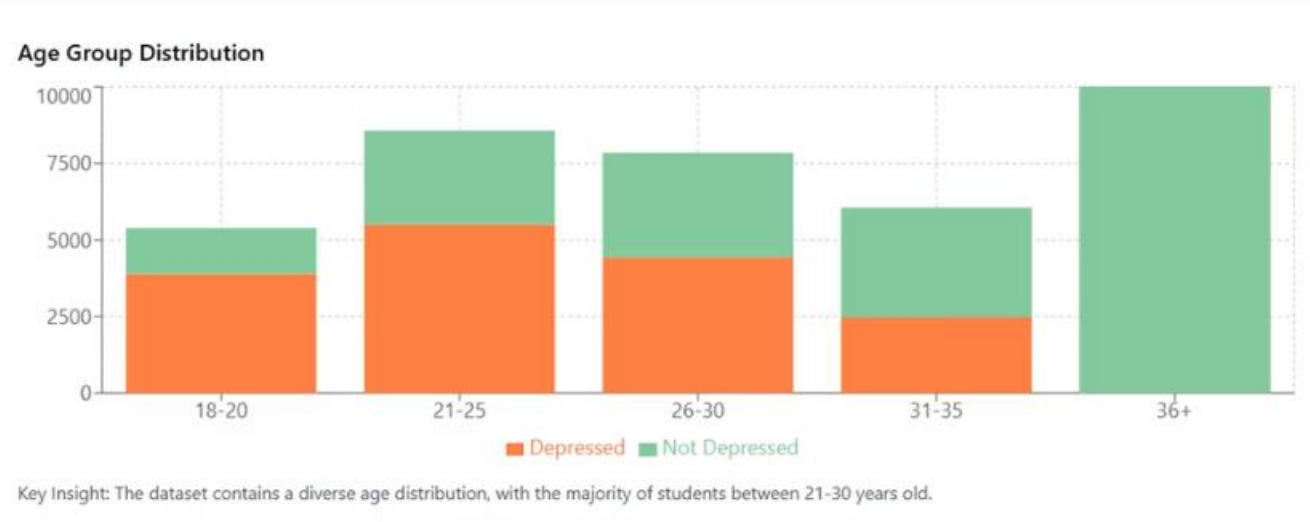
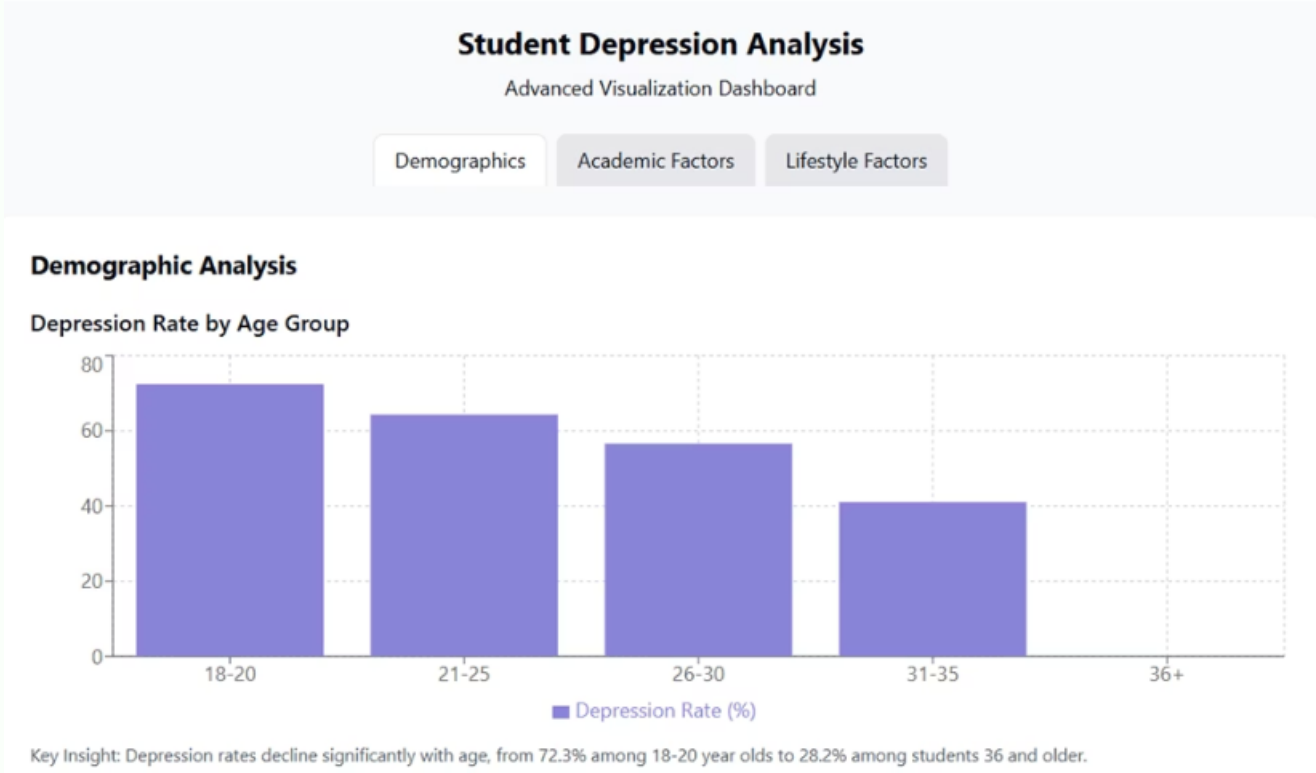
Normalized numerical variables for scale consistency.



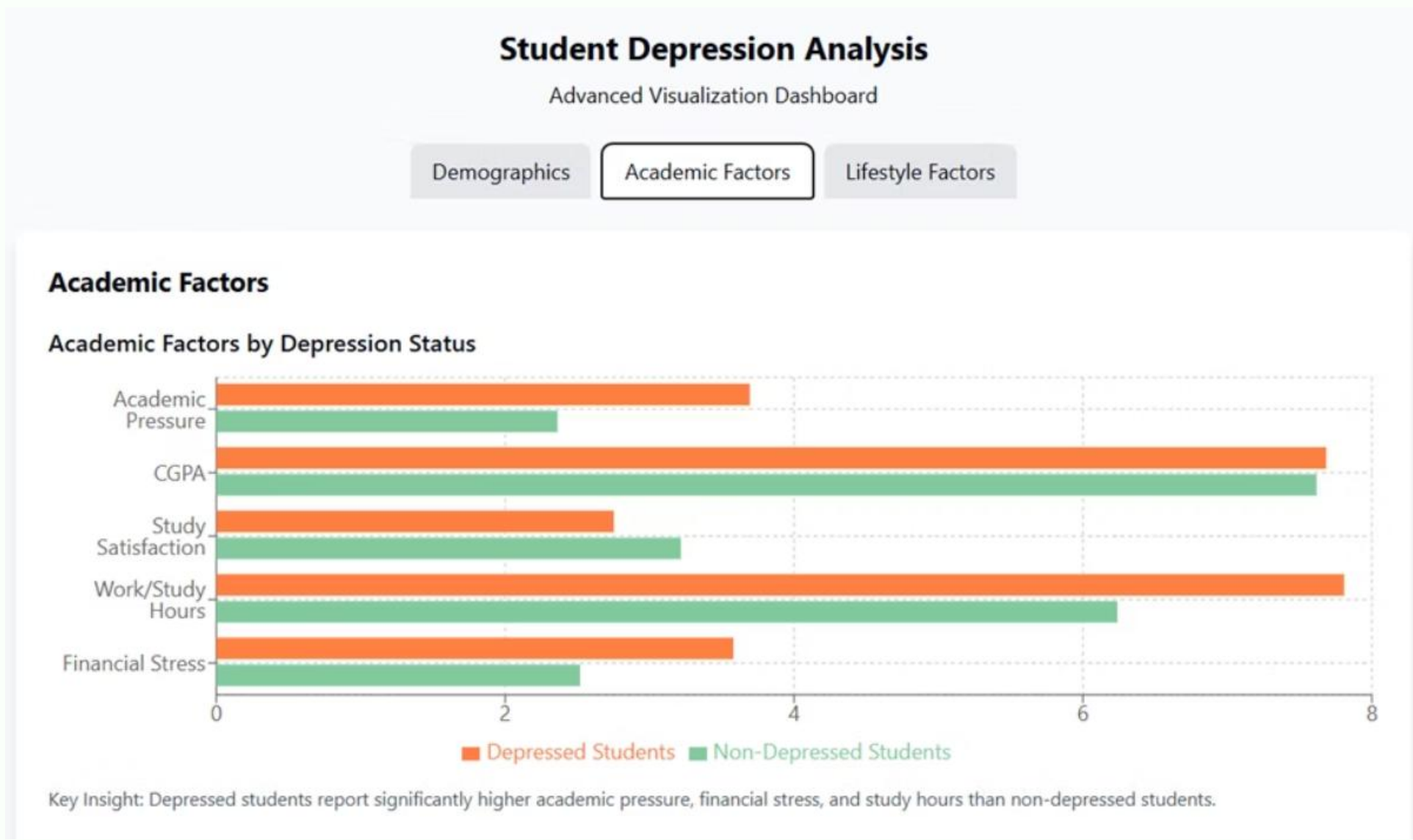
Feature Engineering

- Binary variables encoded to 0/1 format for model compatibility.
- Multi-category features transformed via one-hot encoding.
- "Stress Index" aggregates multiple stress factors numerically.
- Normalized key numeric features like CGPA, Age, and work hours.

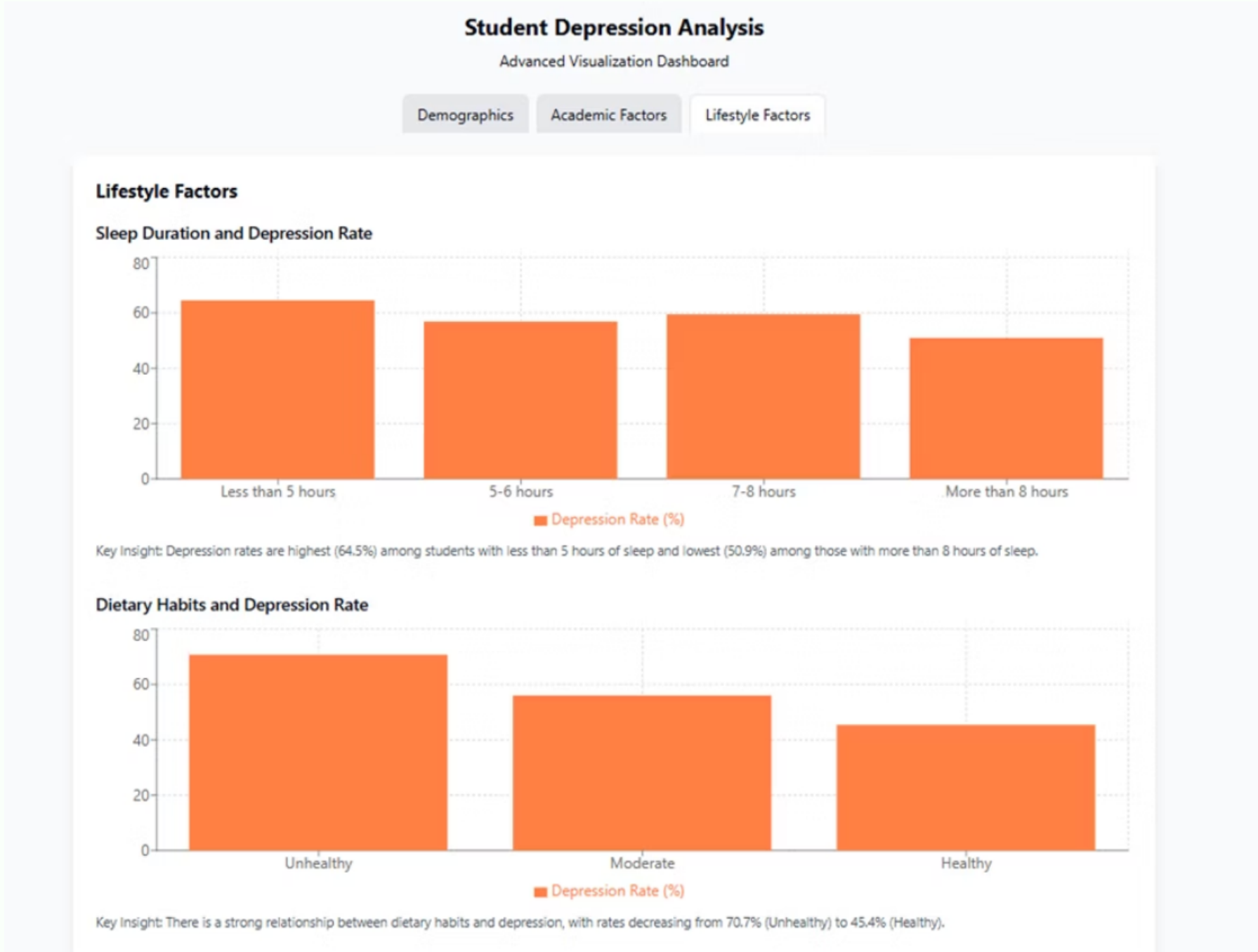
Advanced Visualisation Dashboard: Demographics



Advanced Visualisation Dashboard: Academic Factor



Advanced Visualisation Dashboard: Lifecycle Factor



Exploratory Data Analysis (EDA)



Suicidal Thoughts

Strong indicator linked to depression



Academic Pressure

Major correlation with depression levels



Financial Stress

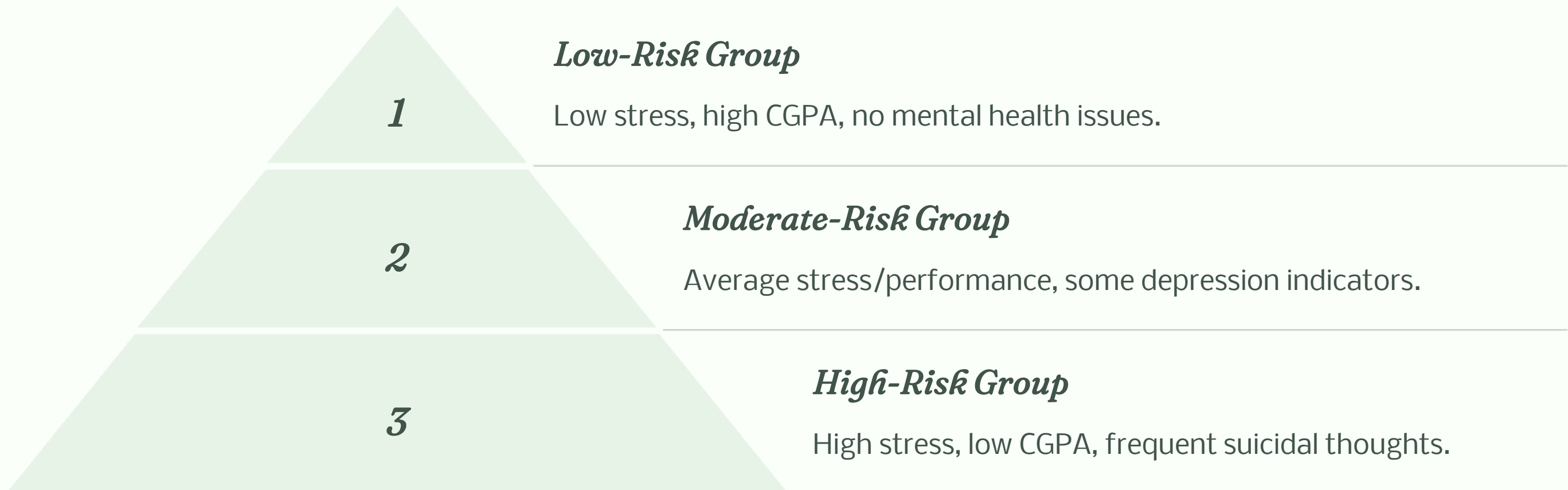
Significantly impacts student mental health



Additional Findings

- Less than 5 hours sleep raises risk
- Higher depression in female students

Clustering Analysis: Student Profiles



Key Takeaways and Next Steps

Actionable Insights

Identify students at risk early using our model.

Targeted Interventions

Develop support programs based on risk profiles.

Further Research

Integrate additional data sources and refine models.

Collaboration

Partner with universities for deployment and feedback.



An illustration on the left side of the slide shows a person in a teal jacket standing at a desk, gesturing towards a large, glowing teal light cone that emanates from a lamp above. Inside the light cone, there are faint, stylized data visualizations including a line graph, a bar chart, and a pie chart. On the desk, there are two potted plants, one small and one large, and some books or papers. The background is a solid teal color.

Model Development

Logistic Regression

Simple baseline with interpretable coefficients.

- Clear feature importance
- Performance baseline

Decision Tree

Captures non-linear variable relationships.

- Visual pathways
- Handles complex interactions

Random Forest

Ensemble approach for high accuracy.

- Reduces overfitting risk
- Improves generalization

Key variables: Suicidal Thoughts, Academic Pressure, Sleep Duration.

Evaluation Metrics

Our depression prediction models were evaluated using key metrics to ensure reliable screening for at-risk students.



Accuracy

Overall percentage of correct predictions, giving a general performance snapshot.



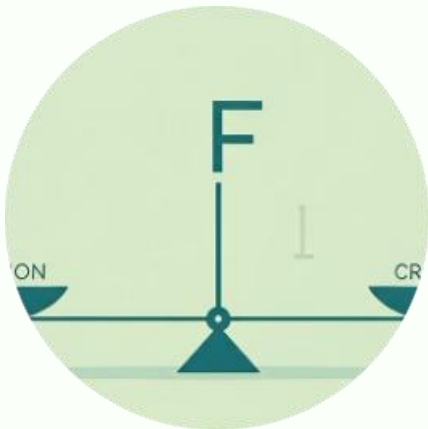
Precision

The proportion of predicted depressed cases that were truly depressed, reducing false alarms.



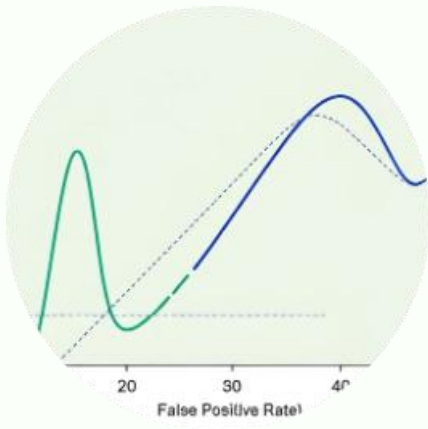
Recall

How many actual depressed students were correctly identified, minimizing missed cases.



F1-Score

Balances precision and recall, representing the harmonic mean of the two.



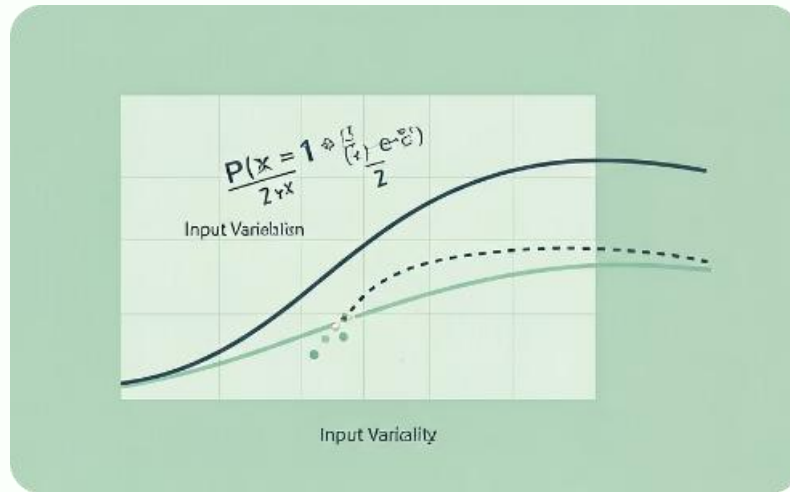
ROC-AUC

Measures model's ability to differentiate between depressed and non-depressed students, assessing threshold quality.

Recall was prioritized to avoid overlooking students at risk during screening.

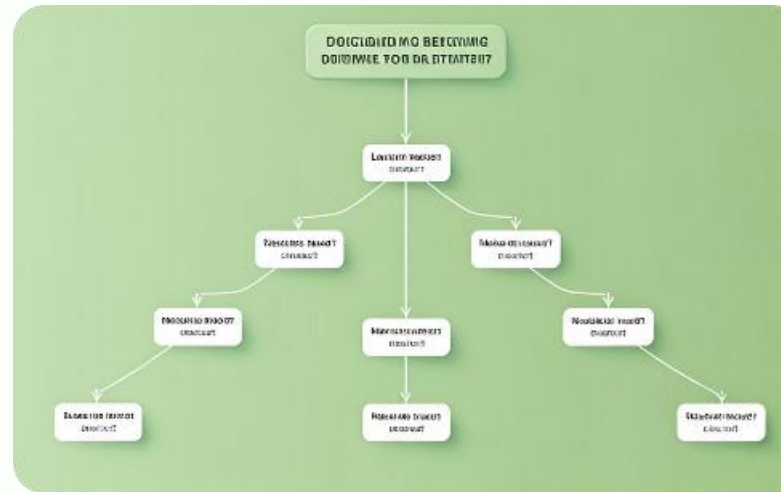
Results: Model Performance Comparison

Our machine learning models progressively improved in detecting depression risk factors among students, highlighted by their accuracy and capabilities.



Logistic Regression

- Accuracy: 78%
- ROC-AUC: 0.82
- Interpretable model with clear feature importance



Decision Tree

- Accuracy: 80%
- ROC-AUC: 0.84
- Provides rule-based insights and captures non-linearities



Random Forest

- Accuracy: 84%
- ROC-AUC: 0.87
- Best overall, robust and reduces overfitting

Common top predictors in all models are Suicidal Thoughts, Academic Pressure, and Sleep Duration.

Feature Importance: Top Predictors



Suicidal Thoughts

Strongest predictor
increasing depression risk



Academic Pressure

High correlation with
depression scores



Financial Stress

Elevates mental health risk
substantially



Sleep Duration

Less than 5 hours linked to
higher depression

Limitations and Ethical Considerations



Self-Reporting Bias

Students may under-report symptoms or exaggerate protective factors.



Missing Variables

Key factors like social media, family dynamics, and culture are not captured.



Population Constraints

Findings may not extend beyond the studied student demographic.



Privacy Safeguards

Careful handling needed to prevent stigmatization from mental health predictions.



Support Requirement

Prediction tools must link to proper intervention and support resources.

Recommendations and Future Work



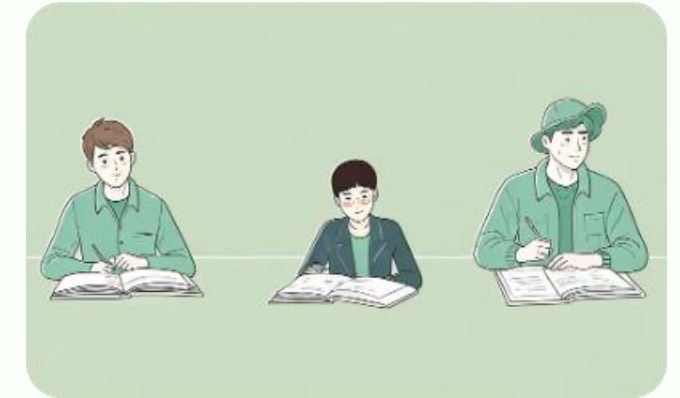
Incorporate behavioral data such as mobile usage and real-time sleep tracking.



Partner with psychologists to develop safe and effective intervention strategies.



Use models on longitudinal datasets to predict future depression risks.



Expand research to various cultures and education systems for wider impact.

Conclusion

Key Depression Predictors

Identified crucial risk factors in students

Top Model: Random Forest

Achieved 84% accuracy in detection

Distinct Risk Profiles

Three groups discovered through clustering

Targeted Support Programs

Helps institutions design focused interventions

References and Acknowledgement

- American College Health Association. *National College Health Assessment*, 2024.
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Thank You!